

AI-Assisted Lending Intelligence Module (ALIM)

Functional Requirements Document (FRD)

Document Control

Attribute	Value
Project Name	AI-Assisted Lending Intelligence Module (ALIM)
Document Type	Functional Requirements Document
Version	1.0
Prepared By	Business Analysis & Solution Architecture Team
Target System	Existing Enterprise Lending Platform
Deployment Model	Modular Service within Existing Monorepo
Architecture Style	Event-Driven Microservices
Intended Users	Underwriters, Operations Teams, Risk Teams, Compliance Teams

1. Executive Summary

The AI-Assisted Lending Intelligence Module (ALIM) is a new enterprise module to be integrated into the existing lending platform. The objective of the module is to augment underwriting and operational workflows using AI-assisted contextual analysis, semantic retrieval, document intelligence, and explainable recommendations.

The system is designed to improve:

- underwriting efficiency
- operational turnaround time
- policy interpretation
- document understanding
- contextual risk analysis
- auditability of AI-assisted operations

The module is not intended to replace underwriting or automate lending decisions. All final decisions remain under deterministic business rules and authorized human users.

All AI-assisted capabilities shall execute within enterprise-controlled infrastructure using approved offline or privately hosted models to ensure confidentiality and regulatory compliance.

2. Business Objectives

2.1 Primary Objectives

- Reduce manual underwriting effort
- Improve loan review turnaround time
- Provide contextual AI-assisted underwriting observations
- Enable semantic policy search and retrieval
- Improve document understanding capabilities
- Establish explainable and auditable AI integration within lending workflows

2.2 Secondary Objectives

- Create reusable enterprise AI infrastructure
- Enable future intelligent workflow expansion
- Standardize AI interaction patterns within the organization
- Provide architecture foundation for future AI-assisted modules

3. Current System Challenges

Area	Existing Problem
Underwriting	Manual document review and interpretation
Policies	Operational teams manually search policy documents
Documents	Unstructured customer documents require manual effort
Risk Analysis	Limited contextual analysis capability
Compliance	Lack of AI explainability and traceability
Operations	Delays in loan processing and review

4. Scope

4.1 In Scope

The following capabilities are included in Phase-1 and Phase-2 implementation:

Document Intelligence

- document ingestion
- text extraction
- semantic chunking
- embedding generation
- vector storage
- support for offline/self-hosted AI inference
- enterprise-confidential AI processing

AI-Assisted Loan Review

- loan summarization
- contextual risk observations
- underwriting assistance
- anomaly indications

Policy Intelligence

- semantic policy retrieval
- policy question answering
- contextual policy lookup

Validation Intelligence

- missing document validation
- contextual document completeness analysis
- rule-driven document verification

Explainability & Audit

- prompt logging

- retrieved context logging
- AI response tracking
- model metadata tracking
- confidence indicators

Workflow Integration

- human-in-loop review lifecycle
- underwriting review stages
- AI review workflow states

4.2 Out of Scope

The following capabilities are explicitly excluded from current implementation:

- autonomous loan approval
- automated loan rejection
- customer-facing chatbot
- fraud prediction ML models
- bureau integration
- real-time streaming ingestion
- multilingual AI support
- voice-based AI interactions
- AI-generated loan eligibility decisions
- self-learning underwriting systems

5. Stakeholders

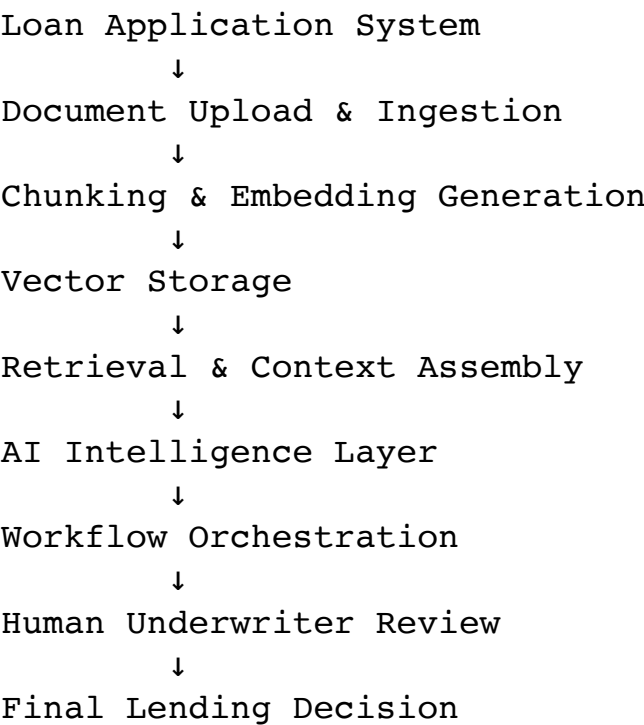
Stakeholder	Responsibility
Underwriters	Review AI-assisted observations
Operations Team	Manage document workflows
Risk Team	Analyze contextual risk insights

Compliance Team	Review audit and explainability records
Technology Team	System implementation and support
Product Team	Feature governance and roadmap
Security Team	Security and compliance governance

6. User Roles

Role	Description
UNDERWRITER	Reviews AI-assisted recommendations
OPERATIONS_USER	Uploads and manages documents
RISK_ANALYST	Reviews contextual risk insights
COMPLIANCE_USER	Reviews audit and explainability logs
ADMIN	Configures system settings
SYSTEM	Internal asynchronous processing actor

7. High-Level Functional Architecture



8. Functional Requirements

FR-1 Document Ingestion

Description

The system shall ingest loan-related documents and convert them into semantically searchable content.

Inputs

Field	Mandatory	Description
loanId	Yes	Unique loan identifier
customerId	Yes	Unique customer identifier
documentType	Yes	Type of uploaded document
file	Yes	Uploaded document file
uploadedBy	Yes	User uploading the file

Supported Document Types

- LOAN_APPLICATION
- SALARY_SLIP
- BANK_STATEMENT
- FORM_16
- ID_PROOF
- ADDRESS_PROOF
- POLICY_DOCUMENT
- UNDERWRITER_NOTE

Processing Rules

The system shall:

1. validate uploaded file format
2. extract textual content
3. generate semantic chunks

4. create embeddings
5. store vector representations
6. persist document metadata
7. trigger ingestion completion event

Outputs

Output	Description
ingestionId	Unique ingestion reference
ingestionStatus	SUCCESS / FAILED
chunkCount	Total generated chunks
embeddingStatus	Embedding generation result

Business Rules

Rule ID	Rule
BR-ING-001	Unsupported file formats shall be rejected
BR-ING-002	Empty documents shall not be ingested
BR-ING-003	Each chunk must contain source metadata
BR-ING-004	Embeddings must be associated with loanId

FR-2 Semantic Retrieval

Description

The system shall retrieve semantically relevant content using vector similarity search.

Inputs

Field	Description
query	Natural language query
loanId	Optional loan context

customerId	Optional customer context
topK	Maximum retrieval count

Functional Flow

1. generate query embeddings
2. perform similarity search
3. apply metadata filters
4. retrieve relevant chunks
5. return ranked results

Outputs

Output	Description
retrievedChunks	Ranked semantic chunks
similarityScore	Similarity confidence
sourceReferences	Source metadata

FR-3 AI-Assisted Loan Summarization

Description

The system shall generate contextual loan summaries for underwriting assistance.

Inputs

Field	Description
loanId	Loan identifier
customerId	Customer identifier

Functional Flow

1. retrieve relevant documents
2. assemble contextual prompt

- 3. invoke LLM service
- 4. generate structured summary
- 5. persist AI interaction metadata

Output Sections

The generated summary shall contain:

- borrower overview
- income observations
- liabilities summary
- repayment observations
- document consistency observations
- contextual risk indicators
- recommendation notes

Sample Output

```
{
  "borrowerProfile": "Salaried employee with stable income
profile",
  "monthlyIncome": 125000,
  "riskIndicators": [
    "Multiple high-value cash deposits observed"
  ],
  "manualReviewRecommended": true
}
```

Business Rules

Rule ID	Rule
BR-SUM-001	AI output shall remain advisory only
BR-SUM-002	Generated summaries must include evidence references
BR-SUM-003	Summary generation shall be auditable

FR-4 AI-Assisted Underwriting Review

Description

The system shall provide contextual underwriting assistance based on loan documents and policies.

Example Queries

- Why might this loan require manual review?
- Identify repayment risks for this applicant
- Are there inconsistencies in income evidence?
- Highlight operational concerns

Functional Flow

1. retrieve contextual documents
2. retrieve relevant policy content
3. build AI context package
4. invoke LLM service
5. generate underwriting observations
6. persist audit trail

Outputs

Output	Description
observations	AI-generated review observations
confidenceIndicator	LOW / MEDIUM / HIGH
sourceReferences	Supporting document references
reviewRecommendation	Suggested next action

Business Rules

Rule ID	Rule
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BR-REV-001	AI recommendations shall not auto-approve loans
BR-REV-002	All outputs must contain supporting evidence
BR-REV-003	AI confidence score must be displayed

FR-5 Policy Question Answering

Description

The system shall provide semantic retrieval and contextual responses over lending policy documents.

Inputs

Field	Description
question	User question
productType	Lending product context

Functional Flow

1. retrieve policy chunks
2. rank relevant content
3. assemble contextual response
4. generate answer using LLM
5. attach source references

Outputs

Output	Description
answer	Generated contextual answer
sourceChunks	Supporting policy references
policyVersion	Referenced policy version

Business Rules

Rule ID	Rule
BR-POL-001	Only approved policy documents may be queried
BR-POL-002	Responses must include policy references
BR-POL-003	Expired policies must not be retrieved

FR-6 Missing Document Validation

Description

The system shall validate document completeness using hybrid rule-based and AI-assisted validation.

Functional Flow

Rule-Based Validation

The system shall:

- validate mandatory document presence
- validate document type mapping
- validate product-specific document requirements

AI-Assisted Validation

The system shall:

- identify semantically incomplete documents
- detect mismatched document content
- identify contextual anomalies

Outputs

Output	Description
missingDocuments	Missing mandatory documents

invalidDocuments	Semantically inconsistent documents
contextualObservations	AI-assisted observations

Business Rules

Rule ID	Rule
BR-DOC-001	Mandatory documents shall be validated via rules
BR-DOC-002	AI observations shall remain advisory
BR-DOC-003	Validation results must be auditable

FR-7 Explainability & Audit Logging

Description

The system shall maintain complete traceability of AI-assisted interactions.

Audit Attributes

The system shall store:

- request identifiers
- prompts
- retrieved chunks
- model versions
- embeddings metadata
- AI responses
- confidence indicators
- user actions
- timestamps

Audit Search Capabilities

Users shall be able to:

- search AI interactions
- retrieve historical prompts
- review source references
- inspect retrieved evidence

Business Rules

Rule ID	Rule
BR-AUD-001	All AI responses must be traceable
BR-AUD-002	Audit records shall be immutable
BR-AUD-003	Audit retention policy shall be configurable

FR-8 Workflow Integration

Description

The system shall integrate AI review lifecycle with lending workflow stages.

Workflow States

State	Description
AI_REVIEW_PENDING	Awaiting AI analysis
AI_REVIEW_COMPLETED	AI analysis completed
UNDERWRITER_REVIEW_PENDING	Awaiting manual review
MANUAL_OVERRIDE	Human override performed
FINAL_DECISION_COMPLETED	Lending decision finalized

Functional Flow

1. loan submitted
2. documents ingested
3. AI review triggered

4. AI response generated
5. underwriter review initiated
6. final workflow transition executed

9. Event Model

The system shall support event-driven asynchronous processing.

Domain Events

Event Name	Description
LoanSubmitted	Loan application created
DocumentUploaded	Document uploaded
DocumentIngested	Document processing completed
EmbeddingGenerated	Embedding generation completed
AiReviewRequested	AI review initiated
AiReviewCompleted	AI review completed
ManualReviewRequired	Human review required
UnderwriterApproved	Underwriter approved loan
LoanRejected	Loan rejected

10. Non-Functional Requirements

NFR-1 Performance

Function	Target SLA
Document ingestion	< 5 seconds
Semantic retrieval	< 2 seconds
AI summarization	< 8 seconds
Policy Q&A	< 5 seconds

NFR-2 Scalability

The system shall:

- support horizontal scaling
- support asynchronous processing
- support stateless service deployment
- support distributed vector search

NFR-3 Security

The system shall:

- enforce role-based access control
- encrypt sensitive data
- mask PII data where required
- support API authentication
- maintain audit compliance

NFR-4 Availability

The target availability shall be 99.5%.

NFR-5 Explainability

Every AI-generated response shall contain:

- source references
- confidence indicator
- retrieved evidence metadata
- generation timestamp

NFR-6 Observability

The system shall provide:

- request tracing

- AI latency metrics
- vector search metrics
- token usage metrics
- error monitoring
- workflow monitoring

NFR-7 Enterprise AI Confidentiality

The system shall ensure that:

- no customer data is transmitted to public AI providers
- all inference workloads execute within enterprise-controlled infrastructure
- prompts and embeddings remain within organization boundaries
- model access is centrally governed
- deployment supports air-gapped environments

11. Proposed Technical Architecture

Existing Monorepo

```
backend/  
├── common/  
├── ai-core/  
├── ingestion/  
├── lending/  
└── app/
```

Recommended Expanded Architecture

```
backend/  
├── common/  
├── ai-core/  
├── ingestion/  
├── lending/  
├── workflow/  
├── rule-engine/  
└── audit/
```

—	policy/
—	document-intelligence/
—	orchestration/
—	integration/
—	app/

12. Recommended Technology Stack

Area	Recommendation
Workflow Engine	Camunda 8 / Temporal
Event Streaming	Kafka
AI Framework	Spring AI
Vector Database	ChromaDB / PGVector
Rule Engine	Drools
Database	PostgreSQL
OCR	AWS Textract / Tesseract
Monitoring	Prometheus + Grafana
Tracing	OpenTelemetry
Security	Spring Security + JWT

13. Integration Requirements

System	Purpose
Existing Lending System	Loan workflow integration
Document Storage	Source document management
Identity Provider	Authentication and authorization
Monitoring Platform	Metrics and observability
Workflow Engine	Human review orchestration

14. Risks & Considerations

Risk	Mitigation
AI hallucinations	Evidence-backed responses
Incorrect recommendations	Human approval mandatory

Policy inconsistency	Version-controlled policies
Compliance concerns	Explainability and audit logging
High latency	Async orchestration and caching

15. Assumptions

- Existing lending platform already manages loan lifecycle
- Existing authentication framework is available
- Existing document upload infrastructure exists
- AI APIs are externally available
- Loan identifiers are globally unique

16. Future Roadmap

Phase 2

- OCR support
- workflow orchestration
- hybrid retrieval
- advanced audit dashboards
- confidence-based routing

Phase 3

- MCP integration
- agentic workflow orchestration
- anomaly scoring models
- bureau integration
- advanced operational intelligence

17. Final Architectural Positioning

The AI-Assisted Lending Intelligence Module (ALIM) is an enterprise-grade intelligence augmentation layer designed to integrate contextual AI capabilities into regulated lending workflows while maintaining deterministic governance, explainability, compliance traceability, and human-controlled decisioning.